

Technical Document #188: Retrieval Augmented Generation - Comprehensive Overview

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Context window optimization selects the most relevant information to fit within the LLM's context limit. It maximizes information density.

Hallucination detection identifies when LLMs generate factually incorrect information. It is critical for maintaining trust in RAG systems.

Semantic search finds documents based on meaning rather than exact keywords. It uses embedding similarity to match queries with relevant content.

Retrieval-Augmented Generation (RAG) combines information retrieval with text generation. It grounds LLM responses in factual, retrieved documents.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Key Takeaways:

- Semantic search finds documents based on meaning rather than exact keywords.
- Query decomposition breaks complex queries into simpler sub-queries.
- Re-ranking models score retrieved documents for relevance to the query.